Continuous-Time Linear Time-Invariant (LTI) System

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1 Continuous-Time Linear Time-Invariant (LTI) System

I Definition

i Causal continuous-time linear time-invariant (LTI) system

A causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation, given initial condition $y(0) = 0$. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$ and is such that $h(t) = 0$ for all $t < 0$, that

$$y(t) = (x * h)(t) \coloneqq \int_0^t x(\tau)h(t-\tau)\,\mathrm{d}\tau\,.$$

Applying unilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is a rational function called transfer function (轉移函數). Unless otherwise specified, we will discuss casual continuous-time linear time-invariant (LTI) system below.

ii Non-causal continuous-time linear time-invariant (LTI) system

A non-causal continuous-time linear time-invariant (LTI) system of a variable $t \in \mathbb{R}$ is a system that accept an input (distribution) (often called function) $x(t)$ and produces a response (distribution) (often called function) $y(t)$ that is the solution of a linear autonomous ODE whose forcing term is $x(t)$, called governing equation. It can be completely characterized by a distribution called impulse response (function) $h(t)$, which is the output when the input is the Dirac delta function $\delta(t)$, that

$$y(t) = (x * h)(t) \coloneqq \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,\mathrm{d}\tau\,.$$

Applying bilateral Laplace transform

$$Y(s) = X(s) \cdot H(s),$$

where $H(s)$ is a rational function called transfer function.

iii Gain

$|H(s)|$ is called gain (增益), or attenuation (衰減量) if < 1.

iv Order

The number of poles of $H(s)$ is called the order of the system.

II Response

i Poles and zeros

Since $H(s)$ is rational, a complex conjugate of a pole is a pole of same order, and a complex conjugate of a zero is a zero of same order.

A pole–zero plot or pole–zero constellation is a graphical representation showing poles and zeros of a transfer function in the complex plane.

ii Transient response

A response component that converges to an equilibrium as t gets large. Such converged state is called steady state or transient state.

iii Time constant

For a response in the form

$$y(t) = A + \sum_{j=1}^{n} B_j e^{-\frac{t}{\tau_j}}$$

with $A \in \mathbb{R}$, $B_j \neq 0$, and $\tau_j > 0$.

Each τ_j is called a time constant, and the system is called a n-time-constant system, e.g., single-time-constant (STC), two-time-constant (TTC).

For each term

$$y_j(t) = B_j e^{-\frac{t}{\tau_j}}$$

if $B_j > 0$, we typically call that

- $y_j(t)$ decays to 0 about $5\tau_j$.
- The falling time or pulse decay time is defined as the time $y_j(t)$ changes from $0.9B_j$ to $0.1B_j$, which is $\ln 9\tau_j \approx 2.2\tau$.

if $B_j < 0$, we typically call that

- $y_j(t)$ grows to B_j about $5\tau_j$.
- The rising time or pulse grow time is defined as the time $y_j(t)$ changes from $0.1B_j$ to $0.9B_j$, which is $\ln 9\tau_j \approx 2.2\tau$.

iv Natural frequency

For a pair of conjugate poles $\pm i\omega$ of $H(s)$, ω is called a natural frequency of the system.

v Resonance

Resonance occurs when the input and transfer function have common poles on the imaginary axis. Such natural frequency is called resonant/resonance frequency.

III Stability

i Asymptotically stable or bounded-input bounded-output (BIBO) stable

A LTI system is called asymptotically stable or bounded-input bounded-output stable iff bounded input implies bounded response, iff all poles of it are on the left half-plane iff $h(t)$ is absolutely integrable.

ii (Lyapunov) stable

A LTI system is called (Lyapunov) stable iff there is no pole of it on the right half-plane and no multiple pole of it on the imaginary axis iff $h(t)$ is bounded.

iii Marginally stable

A LTI system is called marginally stable iff it is stable but not asymptotically stable.

iv Unstable

A LTI system is called unstable iff it is not stable.

IV Frequency response

i Frequency response

- $H(i\omega)$ is called frequency response,
- $a(\omega) \coloneqq |H(i\omega)|$ is called magnitude/amplitude response/ratio or gain,
- $\theta(\omega) \coloneqq \arg(H(i\omega))$ (also denoted as $\varphi(\omega)$) is called phase (angle) response/shift.

ω is called angular frequency and $f \coloneqq \dfrac{\omega}{2\pi}$ is called non-angular frequency. Both can be called frequency and can be differentiated with symbol, and angular frequency is used below.

ii Steady-state sinusoidal response

Assume the system is stable with m LHP poles $a_1, \dots, a_n$ and n pure imaginary poles $b_1, \dots, b_n$ and that $H(s) = O(1)$. When the input is $x(t) = e^{i\omega t}$, $X(s) = \dfrac{1}{s - i\omega}$, the response is

$$Y(s) = \frac{H(s)}{s - i\omega}.$$

$$y(t) = \frac{1}{2\pi i} \lim_{T\to\infty} \int_{\gamma - iT}^{\gamma + iT} Y(s) e^{st} \,\mathrm{d}s\,, \quad \gamma > 0.$$

Close the contour from $\gamma - iT$ to $\gamma + iT$ with a semicircular contour C_T

$$C_T = \{\gamma + iTe^{i\theta} : \theta \in [0, \pi]\}$$

to form a closed contour Γ_T.

$$\begin{aligned}\left|\int_{C_T} Y(s)e^{st}\,\mathrm{d}s\right| &= \left|\int_{C_T} Y(s)e^{(\gamma-T\sin\theta)t}e^{i\Im(s)t}\,\mathrm{d}s\right| \\ &= \int_0^\pi \left|Y(\gamma+iTe^{i\theta})e^{(\gamma-T\sin\theta)t}(-Te^{i\theta})\right|\mathrm{d}\theta \\ &\le Te^{\gamma t}\int_0^\pi \left|Y(\gamma+iTe^{i\theta})\right|e^{-T\sin\theta t}\,\mathrm{d}\theta\end{aligned}$$

$$Y(s) = O\left(\frac{1}{s}\right)$$

Suppose for some $M > 0$, for all sufficiently large $|s|$,

$$|Y(s)| \le \frac{M}{|s|}$$

For all sufficiently large T,

$$\left|\gamma + iTe^{i\theta}\right| \ge T - \gamma$$

$$\left|Y(\gamma+iTe^{i\theta})\right| \le \frac{M}{T-\gamma} = O\left(\frac{1}{T}\right)$$

$$Te^{\gamma t}\int_0^\pi \left|Y(\gamma+iTe^{i\theta})\right|e^{-T\sin\theta t}\,\mathrm{d}\theta = O\left(\int_0^\pi e^{-T\sin\theta t}\,\mathrm{d}\theta\right)$$

$$\int_0^\pi e^{-T\sin\theta t}\,\mathrm{d}\theta = 2\int_0^{\frac{\pi}{2}} e^{-T\sin\theta t}\,\mathrm{d}\theta$$

$$\sin\theta \ge \frac{2\theta}{\pi}$$

$$e^{-T\sin\theta t} \le e^{-\frac{2T\theta t}{\pi}}$$

$$\begin{aligned}2\int_0^{\frac{\pi}{2}} e^{-T\sin\theta t}\,\mathrm{d}\theta &\le 2\int_0^{\frac{\pi}{2}} e^{-\frac{2T\theta t}{\pi}}\,\mathrm{d}\theta \\ &= 2\frac{\pi}{2Tt}\left\{e^{-\frac{2T\theta t}{\pi}}\right\}\Big|_0^{\frac{\pi}{2}} \\ &= \frac{\pi}{Tt}\left(1-e^{-Tt}\right) \\ &\le \frac{\pi}{Tt}\end{aligned}$$

$$2\int_0^{\frac{\pi}{2}} e^{-T\sin\theta t}\,\mathrm{d}\theta = O\left(\frac{1}{T}\right)$$

$$Te^{\gamma t}\int_0^\pi \left|Y(\gamma+iTe^{i\theta})\right|e^{-T\sin\theta t}\,\mathrm{d}\theta = O\left(\frac{1}{T}\right)$$

Thus,

$$\lim_{T\to\infty}\int_{C_T} Y(s)e^{st}\,\mathrm{d}s = 0$$

$$y(t) = \frac{1}{2\pi i}\lim_{T\to\infty}\int_\Gamma Y(s)e^{st}\,\mathrm{d}s$$

Since $H(s)$ is meromorphic, by residue theorem,

$$y(t) = \mathrm{Res}\left(Y(s)e^{st}, i\omega\right) + \sum_{k=1}^{m}\mathrm{Res}\left(Y(s)e^{st}, a_k\right) + \sum_{k=1}^{n}\mathrm{Res}\left(Y(s)e^{st}, b_k\right)$$

Use Laurent series:

$$\operatorname{Res}\big(Y(s)e^{st}, z\big) = \lim_{s\to z}(s-z)Y(s)e^{st} = e^{zt}\lim_{s\to z}(s-z)Y(s)$$

$\sum_{k=1}^{m}\operatorname{Res}\big(Y(s)e^{st}, a_k\big)$ is transient response.

- If $n = 0$, the system is asymptotically stable, and

  $$\operatorname{Res}\big(Y(s)e^{st}, i\omega\big) = H(i\omega)e^{i\omega t}$$

  is steady-state sinusoidal response with amplitude $a(\omega)$ and phase shift (output leading input as positive) $\theta(\omega)$.

- If $n > 0$, the system is marginally stable, and:
  - If $i\omega$ is not a pole,

    $$\operatorname{Res}\big(Y(s)e^{st}, i\omega\big) + \sum_{k=1}^{n}\operatorname{Res}\big(Y(s)e^{st}, b_k\big) = H(i\omega)e^{i\omega t} + \sum_{k=1}^{n} e^{b_k t}\lim_{s\to b_k}\big(s-b_k\big)Y(s)$$

    is steady-state sinusoidal response and the second term is not transient.
  - If $i\omega$ is a pole, it is resonant response that grows to infinity over time.

iii Logarithmic scale and decibel (dB)

Frequency response are often expressed in logarithmic scale:

- Magnitude response, input, and output can be expressed in decibel (dB) as $g(\omega) \coloneqq C\log_{10}|H(i\omega)|$, $C\log_{10}|x|$, and $C\log_{10}|y|$ with some constant C. $C = 20$ is used here.
- Changing by 2 times is called an octave, oct or 8ve, and changing by 10 times is called a decade or dec. k dec^{-1} equals to $\log_{10}(2)$ oct^{-1}. The most common used is 20 dB/dec equals to $20\log_{10}(2) \approx 6$ dB/oct.
- Gradually reducing gain at either end of the frequency range is called roll-off.

iv Midband gain and normalizing

For a transfer function $H(s)$ such that

$$\sup_{\omega} a(\omega) = K \in \mathbb{R}_{>0},$$

K is called midband gain, and normalizing means dividing $H(s)$ by K.

v Bode plot (波德圖)

- Bode magnitude plot or dB gain curve: Magnitude response plotted with frequency in logarithmic scale as horizontal axis and magnitude response (or, normalized magnitude response, where the Bode magnitude plot is called normalized) in dB as vertical axis.

- Bode phase plot or phase curve: Phase response plotted with frequency in logarithmic scale as horizontal axis and phase response as vertical axis.
- Asymptotic Bode plot: The asymptotes of Bode plots for low and high frequency is often interested.

vi Cutoff (截止), break, or corner (angular) frequency

For a frequency response with finite midband gain K, a cutoff frequency ω_c is a frequency such that

$$\left|H(i\omega_c)\right| = \frac{K}{\sqrt{2}}.$$

$20\log_{10}\left(\frac{1}{\sqrt{2}}\right) = -10\log_{10}(2) \approx -3$, thus this frequency is also denoted $\omega_{-3\text{dB}}$ or $\omega_{3\text{dB}}$.

vii Bandwidth

Bandwidth (or -3 dB bandwidth) of a frequency response with finite midband gain is the set of frequencies over which the normalized gain is no less than $\frac{1}{\sqrt{2}}$, or the length of it if it is connected. k dB bandwidth of a frequency response with finite midband gain is the set of frequencies over which the normalized gain in dB is no less than k dB, or the length of it if it is connected. 0 dB bandwidth is also called unity-gain bandwidth.

V Filters

i Low-pass or lowpass (低通) (LP), high-cut, or bass-cut filter (LPF)

A continuous-time linear time-invariant system is a low-pass filter iff

$$\lim_{\omega\to 0} a(\omega) = K$$

$$\lim_{\omega\to \infty} a(\omega) = 0$$

where K is midband gain, also called low-pass gain.

ii High-pass or highpass (高通) (HP), low-cut, or treble-cut filter (HPF)

A continuous-time linear time-invariant system is a normalized high-pass filter iff

$$\lim_{\omega\to 0} a(\omega) = 0$$

$$\lim_{\omega\to \infty} a(\omega) = K$$

where K is midband gain, also called high-pass gain.

iii Band-pass or bandpass (BP) filter (BPF)

A continuous-time linear time-invariant system is a standard normalized band-pass filter iff

$$\lim_{\omega \to 0} a(\omega) = 0$$

$$\lim_{\omega \to \infty} a(\omega) = 0$$

$$\arg_{\omega} \left\{ a(\omega) = \frac{K}{\sqrt{2}} \right\} = \omega_L, \omega_H$$

$$\left| H(\sqrt{\omega_L \omega_H}) \right| = K$$

where K is midband gain, also called band-pass gain, ω_L, ω_H are called band edge, $B = \omega_H - \omega_L$ is bandwidth, and $\omega_0 = \sqrt{\omega_L \omega_H}$ is called central/center frequency.

If $\omega_0 \gg B$, the filter is called narrow.

iv Band-stop or bandstop (BS), band-reject, or notch filter (BSF)

A continuous-time linear time-invariant system is a standard normalized band-stop filter iff

$$\arg_{\omega} \left\{ a(\omega) = \frac{K}{\sqrt{2}} \right\} = \omega_L, \omega_H$$

$$\left| H(\sqrt{\omega_L \omega_H}) \right| = 0$$

where K is midband gain, also called band-stop gain or notch gain, ω_L, ω_H are called band edge, $B = \omega_H - \omega_L$ is bandwidth, and $\omega_0 = \sqrt{\omega_L \omega_H}$ is called central/center frequency.

If $\omega_0 \gg B$, the filter is called narrow.

v Normalized low-pass prototype filter

A normalized low-pass prototype filter is a low-pass filter with cutoff frequency $\omega_c = 1$. Suppose the transfer function of it is $H_P(s)$. A normalized low-pass prototype filter is usually defined with $(a(\omega))^2 = H(i\omega)H(-i\omega)$ or $H(s)$.

To transform an nth-order normalized low-pass prototype filter to an nth-order low-pass filter with cutoff frequency ω_c, use the transfer function

$$H_{LP}(s) = H_P\left(\frac{s}{\omega_c}\right).$$

To transform an nth-order normalized low-pass prototype filter to an nth-order high-pass filter with cutoff frequency ω_c, use the transfer function

$$H_{HP}(s) = H_P\left(\frac{\omega_c}{s}\right).$$

To transform an nth-order normalized low-pass prototype filter to a $2n$th-order standard band-pass filter with cutoff frequency ω_c, use the transfer function

$$H_{BP}(s) = H_P\left(\frac{s^2 + {\omega_0}^2}{Bs}\right).$$

To transform an nth-order normalized low-pass prototype filter to a $2n$th-order standard band-stop filter with cutoff frequency ω_c, use the transfer function

$$H_{BS}(s) = H_P\left(\frac{Bs}{s^2 + {\omega_0}^2}\right).$$

vi Butterworth filter or maximally flat magnitude filter

nth-order Butterworth low-pass prototype filter is

$$(a(\omega))^2 = \frac{1}{1+\omega^{2n}}$$

$$H(s) = \prod_{k=1}^{n} \frac{1}{s - e^{\frac{j(2k+n-1)\pi}{2n}}}$$

For first-order,

$$(a(\omega))^2 = \frac{1}{1+\omega^2}$$

$$H(s) = \frac{1}{s+1}$$

When not specified, a filter of a type usually refers to the lowest-order Butterworth filter of the type.

vii Single-time-constant (STC) filters

ω_c is cutoff frequency, τ is time constant, $|K|$ is gain at $\omega = 0$, and the following form is called standard form.

Low-pass:

$$H(s) = \frac{K}{1+\frac{s}{\omega_c}}$$

$$H(i\omega) = \frac{K}{1+i\frac{\omega}{\omega_c}}$$

$$a(\omega) = \frac{|K|}{\sqrt{1+\left(\frac{\omega}{\omega_c}\right)^2}}$$

$$\theta(\omega) = -\arg\left(\frac{\omega}{\omega_c}\right)$$

$$H(i0) = K$$

$$H(i\infty) = 0$$

$$\tau = \frac{1}{\omega_c}$$

High-pass:

$$H(s) = \frac{K}{1-\frac{s}{\omega_c}}$$

$$H(i\omega) = \frac{K}{1-i\frac{\omega}{\omega_c}}$$

$$a(\omega) = \frac{|K|}{\sqrt{1 - \left(\frac{\omega}{\omega_c}\right)^2}}$$

$$\theta(\omega) = \arg\left(\frac{\omega}{\omega_c}\right)$$

$$H(i0) = 0$$

$$H(i\infty) = K$$

$$\tau = \frac{1}{\omega_c}$$

VI Transfer function factors

i Transfer function factoring

We can factor $H(s)$ into

$$H(s) = G\prod_{k=1}^{m} H_k(s)$$

then

$$a(\omega) = |G|\prod_{k=1}^{m} \left|H_k(i\omega)\right|$$

$$g(\omega) = 20\log_{10}|G| + 20\sum_{k=1}^{m} \log_{10}\left|H_k(i\omega)\right|$$

$$\theta(\omega) = \arg(G) + \sum_{k=1}^{m} \arg\left(H_k(i\omega)\right)$$

If H and all H_k have unity maximum gains, and the cutoff frequency of one H_k is such that all other H_k has approximately unity gain, that frequency is approximately a cutoff frequency of H.

For the factors below, variables after ; are parameters, which are positive real numbers unless otherwise specified. Only positive ω are considered.

ii Reciprocal factor

This factor is not asymptotically stable but marginally stable.

$$H_c(s; W) \coloneqq \frac{W}{s}$$

$$H_c(i\omega; W) = \frac{W}{i\omega}$$

$$a_c(\omega; W) = \frac{W}{\omega}$$

$$g_c(\omega; W) = -20\log_{10}(\omega) + 20\log_{10}(W)$$

$$\theta_c(\omega; W) = -\frac{\pi}{2}$$

Bode magnitude plot has slope -20 dB/dec, y intercept $20\log_{10}(W)$, and x intercept $-\log_{10}(W)$.

Limit:

- For $\omega \gg W$:

$$a_c(\omega) \approx 0$$
$$g_c(\omega) \approx -\infty$$

- For $\omega \ll W$:

$$a_c(\omega) \approx \infty$$
$$g_c(\omega) \approx \infty$$

iii Ramp factor

$$H_r(s; W) \coloneqq \frac{s}{W}$$
$$H_r(i\omega; W) = \frac{i\omega}{W}$$
$$a_r(\omega; W) = \frac{\omega}{W}$$
$$g_r(\omega; W) = 20 \log_{10}(\omega) - 20 \log_{10}(W)$$
$$\theta_r(\omega; W) = \frac{\pi}{2}$$

Bode magnitude plot has slope 20 dB/dec, y intercept $-20 \log_{10}(W)$, and x intercept $\log_{10}(W)$.

Limit and cutoff frequency:

- For $\omega \gg W$:

$$a_r(\omega; W) \approx \infty$$
$$g_r(\omega; W) \approx \infty$$

- For cutoff frequency $\omega = \omega_c = \frac{W}{\sqrt{2}}$:

$$a_r(\omega; W) = \frac{1}{\sqrt{2}}$$
$$g_r(\omega; W) = -10 \log_{10}(2)$$

- For $\omega \ll W$:

$$a_r(\omega; W) \approx 0$$
$$g_r(\omega; W) \approx -\infty$$

iv Low-pass factor

$$H_{LP}(s; \omega_c) \coloneqq \frac{\omega_c}{s + \omega_c}$$
$$H_{LP}(i\omega; \omega_c) = \frac{\omega_c}{i\omega + \omega_c}$$
$$a_{LP}(\omega; \omega_c) = \frac{\omega_c}{\sqrt{\omega^2 + {\omega_c}^2}}$$
$$g_{LP}(\omega; \omega_c) = 20 \log_{10}\left(\omega_c\right) - 10 \log_{10}\left(\omega^2 + {\omega_c}^2\right)$$
$$\theta_{LP}(\omega; \omega_c) = -\arg\left(i\omega + \omega_c\right)$$

Asymptotic Bode plot and cutoff frequency:

- For $\omega \gg \omega_c$:

$$a_{LP}(\omega;\omega_c) \approx \frac{\omega_c}{\omega}$$
$$g_{LP}(\omega;\omega_c) \approx 20\log_{10}(\omega_c) - 20\log_{10}(\omega),$$

which is slant asymptote with slope -20 dB/dec, y intercept $20\log_{10}(\omega_c)$, and x intercept $\log_{10}(\omega_c)$.

$$\theta_{LP}(\omega;\omega_c) \approx 0$$

- For cutoff frequency $\omega = \omega_c$:

$$a_{LP}(\omega;\omega_c) = \frac{1}{\sqrt{2}}$$
$$g_{LP}(\omega;\omega_c) = -10\log_{10}(2)$$
$$\theta_{LP}(\omega;\omega_c) = -\frac{\pi}{4}$$

- For $\omega \ll \omega_c$:

$$a_{LP}(\omega;\omega_c) \approx 1$$
$$g_{LP}(\omega;\omega_c) \approx 0,$$

which is horizontal asymptote $g_{LP} = 0$.

$$\theta_{LP}(\omega;\omega_c) \approx -\frac{\pi}{2}$$

v High-pass factor

$$H_{HP}(s;\omega_c) \coloneqq \frac{s}{s+\omega_c}$$
$$H_{HP}(i\omega;\omega_c) = \frac{i\omega}{i\omega+\omega_c}$$
$$a_{HP}(\omega;\omega_c) = \frac{\omega}{\sqrt{\omega^2+{\omega_c}^2}}$$
$$g_{HP}(\omega;\omega_c) = 20\log_{10}(\omega) - 10\log_{10}\left(\omega^2+{\omega_c}^2\right)$$
$$\theta_{HP}(\omega;\omega_c) = \frac{\pi}{2} - \arg\left(i\omega+\omega_c\right)$$

Asymptotic Bode plot and cutoff frequency:

- For $\omega \gg \omega_c$:

$$a_{LP}(\omega;\omega_c) \approx 1$$
$$g_{HP}(\omega;\omega_c) \approx 0,$$

which is horizontal asymptote $g_{HP} = 0$.

$$\theta_{HP}(\omega;\omega_c) \approx \frac{\pi}{2} - \frac{\pi}{2} = 0$$

- For cutoff frequency $\omega = \omega_c$:

$$a_{HP}(\omega;\omega_c) = \frac{1}{\sqrt{2}}$$
$$g_{HP}(\omega;\omega_c) = -10\log_{10}(2)$$
$$\theta_{HP}(\omega;\omega_c) = \frac{\pi}{4}$$

- For $\omega \ll \omega_c$:

$$a_{HP}(\omega;\omega_c) \approx \frac{\omega}{\omega_c}$$

$$g_{HP}(\omega;\omega_c) \approx 20\log_{10}(\omega) - 20\log_{10}(\omega_c),$$

which is slant asymptote with slope 20 dB/dec, y intercept $-20\log_{10}(\omega_c)$, and x intercept $\log_{10}(\omega_c)$.

$$\theta_{HP}(\omega;\omega_c) \approx \frac{\pi}{2}$$

vi Quadratic low-pass factor

$$H_{QD}(s;\omega_0,Q) \coloneqq \frac{{\omega_0}^2}{s^2 + \frac{\omega_0}{Q}s + {\omega_0}^2}, \quad Q \geq \frac{1}{2}$$

$$H_{QD}(i\omega;\omega_0,Q) = \frac{{\omega_0}^2}{-\omega^2 + i\frac{\omega_0}{Q}\omega + {\omega_0}^2}$$

$$a_{QD}(\omega;\omega_0,Q) = \frac{{\omega_0}^2}{\sqrt{\frac{{\omega_0}^2}{Q^2}\omega^2 + \omega^4 + {\omega_0}^4 - 2\omega^2{\omega_0}^2}}$$

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\left(a_{QD}(\omega;\omega_0,Q)\right) = \frac{-\frac{{\omega_0}^4}{Q^2}\omega - 2\omega^3{\omega_0}^2 + 2\omega{\omega_0}^4}{\left(\frac{{\omega_0}^2}{Q^2}\omega^2 + \omega^4 + {\omega_0}^4 - 2\omega^2{\omega_0}^2\right)^{\frac{3}{2}}}$$

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\left(a_{QD}(\omega;\omega_0,Q)\right)\Big|_{\omega=\omega_p} = 0$$

$$\omega_p = \omega_0\sqrt{1 - \frac{1}{2Q^2}}, \quad Q > \frac{1}{\sqrt{2}}$$

$$\theta_{QD}(\omega;\omega_0,Q) = -\arg\left(-\omega^2 + i\frac{\omega_0}{Q}\omega + {\omega_0}^2\right)$$

- If $Q \geq \frac{1}{2}$, it is a low-pass filter.
- If $Q > \frac{1}{2}$, the maximum of g_{QD} occurs at peak frequency $\omega = \omega_p = \omega_0\sqrt{1 - \frac{1}{2Q^2}}$.
- If $Q = \frac{1}{2}$, at $\omega = \omega_0\sqrt{1 - \frac{1}{2Q^2}} = 0$,

$$a_{QD}(\omega;\omega_0,Q) = \infty$$

$$g_{QD}(\omega;\omega,Q) = \infty$$

- If $Q < \frac{1}{2}$, it is unstable.

Asymptotic Bode plot and center and peak frequencies:

- For $\omega \ll \omega_0, \frac{\omega_0}{Q}$:

$$a_{QD}(\omega;\omega_0,Q) \approx 1$$

$$g_{QD}(\omega;\omega_0,Q) \approx 0,$$

which is horizontal asymptote $g_{QD} = 0$.

$$\theta_{QD}(\omega;\omega_0,Q) \approx 0$$

- For center frequency $\omega = \omega_0$:

$$\left|H_{QD}(i\omega;\omega_0,Q)\right| = Q$$

$$g_{QD}(\omega;\omega_0,Q) = 20\log_{10}(Q)$$

$$\theta_{QD}(\omega;\omega_0,Q) = -\frac{\pi}{2}$$

- If $Q > \frac{1}{2}$, for peak frequency $\omega = \omega_p = \omega_0\sqrt{1-\frac{1}{2Q^2}}$:

$$a_{QD}(\omega;\omega_0,Q) = \frac{{\omega_0}^2}{\sqrt{\frac{{\omega_0}^4}{Q^2} - \frac{{\omega_0}^4}{2Q^4} + {\omega_0}^4 - \frac{{\omega_0}^4}{Q^2} + \frac{{\omega_0}^4}{4Q^4} + {\omega_0}^4 - 2{\omega_0}^4 + \frac{{\omega_0}^4}{Q^2}}} = \frac{Q}{\sqrt{1-\frac{1}{4Q^2}}}$$

$$g_{QD}(\omega;\omega_0,Q) = 20\log_{10}(Q) - 10\log_{10}\left(1-\frac{1}{4Q^2}\right),$$

which are maxima $a_{QD} = \frac{Q}{\sqrt{1-\frac{1}{4Q^2}}}$ and $g_{QD} = 20\log_{10}(Q) - 10\log_{10}\left(1-\frac{1}{4Q^2}\right)$.

- For $\omega \gg \omega_0, \frac{\omega_0}{Q}$:

$$a_{QD}(\omega;\omega_0,Q) \approx \frac{{\omega_0}^2}{\omega^2}$$

$$g_{QD}(\omega;\omega_0,Q) \approx 40\log_{10}(\omega_0) - 40\log_{10}(\omega)$$

which is slant asymptote with slope -40 dB/dec, y intercept $40\log_{10}(\omega_0)$, and x intercept $\log_{10}(\omega_0)$.

$$\theta_{QD}(\omega;\omega_0,Q) \approx -\pi$$

vii Band-pass factor

$$H_{BP}(s;B,\omega_0) \coloneqq \frac{Bs}{s^2 + Bs + {\omega_0}^2}$$

$$H_{BP}(i\omega;B,\omega_0) = \frac{iB\omega}{-\omega^2 + iB\omega + {\omega_0}^2}$$

$$a_{BP}(\omega;B,\omega_0) = \frac{B\omega}{\sqrt{B^2\omega^2 + \omega^4 + {\omega_0}^4 - 2\omega^2{\omega_0}^2}}$$

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\left(a_{BP}(\omega;B,\omega_0)\right) = \frac{B\left(\omega^4 + {\omega_0}^4 - 2\omega^2{\omega_0}^2 + 2\omega^3 - 2\omega{\omega_0}^2\right)}{\left(B^2\omega^2 + \omega^4 + {\omega_0}^4 - 2\omega^2{\omega_0}^2\right)^{\frac{3}{2}}}$$

$$\frac{\mathrm{d}}{\mathrm{d}\omega}\left(a_{BP}(\omega; B, \omega_0)\right)\Big|_{\omega=\omega_0} = 0$$

$$\theta_{BP}(\omega; B, \omega_0) = \frac{\pi}{2} - \arg\left(-\omega^2 + iB\omega + {\omega_0}^2\right)$$

Asymptotic Bode plot and center and peak frequency:

- For $\omega \gg \omega_0, B$:

$$a_{BP}(\omega; B, \omega_0) \approx \frac{B}{\omega}$$

$$g_{BP}(\omega; B, \omega_0) \approx -20\log_{10}(\omega) + 20\log_{10}(B)$$

  which is slant asymptote with slope -20 dB/dec, y intercept $20\log_{10}(B)$, and x intercept $\log_{10}(B)$.

$$\theta_{BP}(\omega; B, \omega_0) \approx -\frac{\pi}{2}$$

- For center and peak frequency $\omega = \omega_0$:

$$a_{BP}(\omega; B, \omega_0) = 1$$

$$g_{BP}(\omega; B, \omega_0) = 0$$

  which are maxima $a_{BP} = 1$ and $g_{BP} = 0$.

$$\theta_{BP}(\omega; B, \omega_0) = 0$$

- For $\omega \ll \omega_0, B$:

$$a_{BP}(\omega; B, \omega_0) \approx \frac{B\omega}{{\omega_0}^2}$$

$$g_{BP}(\omega; B, \omega_0) \approx 20\log_{10}(\omega) + 20\log_{10}(B) - 40\log_{10}\left(\omega_0\right),$$

  which is slant asymptote with slope 20 dB/dec, y intercept $20\log_{10}(B) - 40\log_{10}\left(\omega_0\right)$, and x intercept $2\log_{10}\left(\omega_0\right) - \log_{10}(B)$.

$$\theta_{BP}(\omega; B, \omega_0) \approx \frac{\pi}{2}$$

viii Band-stop factor

$$H_{BS}(s; B, \omega_0) \coloneqq \frac{s^2 + {\omega_0}^2}{s^2 + Bs + {\omega_0}^2}$$

$$H_{BS}(i\omega; B, \omega_0) = \frac{-\omega^2 + {\omega_0}^2}{-\omega^2 + iB\omega + {\omega_0}^2}$$

$$a_{BS}(\omega; B, \omega_0) = \frac{-\omega^2 + {\omega_0}^2}{\sqrt{B^2\omega^2 + \omega^4 + {\omega_0}^4 - 2\omega^2{\omega_0}^2}}$$

$$\theta_{BS}(\omega; B, \omega_0) = \arg\left(\frac{-\omega^2 + {\omega_0}^2}{-\omega^2 + iB\omega + {\omega_0}^2}\right)$$

Asymptotic Bode plot and center frequency:

- For $\omega \gg \omega_0, B$:

$$a_{BS}(\omega; B, \omega_0) \approx 1$$

$$g_{BS}(\omega; B, \omega_0) \approx 0,$$

  which is horizontal asymptote $g_{BS} = 0$.

$$\theta_{BS}(\omega; B, \omega_0) \approx 0$$

- For center frequency $\omega = \omega_0$:

$$a_{BS}(\omega; B, \omega_0) = 0$$

$$g_{BS}(\omega; B, \omega_0) = -\infty$$

$$\theta_{BS}(\omega; B, \omega_0) = \arg(0) = \text{undefined}$$

- For $\omega \ll \omega_0, B$:

$$a_{BS}(\omega; B, \omega_0) \approx 1$$

$$g_{BS}(\omega; B, \omega_0) \approx 0$$

  which is horizontal asymptote $g_{BS} = 0$.

$$\theta_{BS}(\omega; B, \omega_0) \approx 0$$